SCOTT SIEGER

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ANALYST | ADVANCED ANALYTICS

SUMMARY

Data science and analytics professional with 4+ years of experience applying large-scale analytics and predictive models to diverse business problems. Experienced in Python and SQL with hands-on experience using various machine learning techniques and data management tools. Proven track record of translating complex analytical findings into business decisions through collaboration with engineering, product, underwriting, and senior leadership.

SKILLS

Languages & Tools: Python (pandas, NumPy, scikit-learn, seaborn), SQL, SAS
Data Platforms: Snowflake, dbt, Git, VS Code
Machine Learning: XGBoost, Elastic Net, K-Means, PCA, Feature Engineering, Cross Validation, Model Validation
Visualization and Reporting: Tableau, Power BI, PowerPoint

PROFESSIONAL EXPERIENCE

Liberty Mutual — Advanced Analytics Analyst *Jun 2025 – Present*

- Lead statistical validation of catastrophe risk models covering billions in insured exposure, using quantitative testing methodologies to assess performance and business impact.
- Partner with Product, Underwriting, and Engineering teams to translate model outputs into pricing and growth recommendations, and to communicate to technical and non-technical stakeholders.

Liberty Mutual — Senior Associate, Risk Analytics *Jun 2023 – Jun 2025*

- Cleaned and prepared millions of rows of messy intake data using Python, SAS, and Snowflake to support ongoing catastrophe risk model deployment, validation, and monitoring.
- Automated data cleaning and feature engineering workflows, reducing quarterly processing time by over 1 week.

Liberty Mutual — Analyst Development Program *Jun 2022 – Jun 2023*

- Developed and validated XGBoost classification model to predict Good Student Discount eligibility for auto policies. Evaluated performance of model (AUC).
- Applied PCA and K-Means clustering to build a territory segmentation framework using HLDI auto data to support pricing and geographic risk analysis. Provided a better grain for territory analysis across 6 states.
- Built a Tableau dashboard to explain pricing factors of a workers' compensation pricing model to help with model buy-in from stakeholders.
- Translated auto pricing elastic net model from SAS to Python enabling version control and performance tracking.

Liberty Mutual — Data Science Intern *May 2021 – Aug 2021*

- Designed a predictive claims segmentation framework using SQL, Python, and Power BI to support claims analysis.

PROJECTS

Letterboxd Movie Recommendation System **Github**

- Built hybrid recommendation system using MovieLens and Letterboxd review / rating data.
- Trained collaborative filtering and content-based models to recommend movies based on a user's own personal rating data.
- Integrated TMDB API to enrich movies with metadata for the content-based model.

EDUCATION

Boston College - B.A. Economics; Minors: Computer Science & International Studies
Dean's List, 2019 - 2022
MIT Professional Education - Applied Data Science & AI Certificate
Capstone Project